Master Roadmap: Zero to Master in AI / ML

Follow phases in order. Each phase lists what to learn, how long it takes (approx. 2 hrs/day), and a checkpoint project. Other PDFs in this folder explain every topic in detail with code and Tanglish explanation.

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1. How to use this folder [Beginner]

Explanation (English)
Read PDFs in numeric order (00 -> 14). Keep the Excel tracker open and update the Status column. For each topic: read the explanation, run the code, watch one video, then build a mini-project. Revise weekly.

Tanglish (Tamil + English) Vilakkam
Indha folder-a number order-la padinga (00 la irundhu 14 varai). Excel tracker-la Status update pannunga. Ovvoru topic-ukkum: explanation padinga, code run pannunga, oru video paarunga, appuram chinna project pannunga. Vaaram vaaram revise pannunga.

Learn more
YouTube search: how to learn machine learning roadmap

2. Phase 1 - Foundations (Weeks 1-4) [Beginner]

Explanation (English)
Learn Python basics, data structures, OOP, file handling, and the minimum maths: vectors, matrices, derivatives, probability, mean/variance. Tools: Python 3.11+, VS Code, Jupyter, Git/GitHub. Checkpoint: write a small CLI program and push it to GitHub.

Tanglish (Tamil + English) Vilakkam
Mudhalla Python basics, data structures, OOP, file handling, konjam maths (vector, matrix, derivative, probability, mean/variance) kathukkonga. Tools: Python, VS Code, Jupyter, Git. Checkpoint: oru chinna program ezhudhi GitHub-la push pannunga.

Learn more
YouTube search: python for beginners full course

3. Phase 2 - Data Skills (Weeks 5-8) [Beginner]

Explanation (English)
NumPy, Pandas, Matplotlib/Seaborn, SQL basics, data cleaning and Exploratory Data Analysis (EDA). Checkpoint: complete EDA on the Titanic or any Kaggle dataset and publish the notebook.

Tanglish (Tamil + English) Vilakkam
NumPy, Pandas, Matplotlib/Seaborn, SQL basics, data cleaning, EDA. Checkpoint: Titanic maadhiri Kaggle dataset-la EDA panni notebook publish pannunga.

Learn more
YouTube search: pandas numpy matplotlib full course

4. Phase 3 - Classical Machine Learning (Weeks 9-16) [Intermediate]

Explanation (English)
Supervised (regression, classification), unsupervised (clustering, PCA), model evaluation, cross-validation, regularisation, tree ensembles (Random Forest, XGBoost), feature engineering, scikit-learn pipelines. Checkpoint: house-price and churn prediction projects, plus a Kaggle competition entry.

Tanglish (Tamil + English) Vilakkam
Supervised (regression, classification), unsupervised (clustering, PCA), evaluation, cross-validation, regularisation, Random Forest/XGBoost, feature engineering, sklearn pipeline. Checkpoint: house price, churn prediction project + oru Kaggle competition.

Learn more
YouTube search: machine learning full course scikit-learn

5. Phase 4 - Deep Learning (Weeks 17-26) [Intermediate]

Explanation (English)
Neural networks from scratch, backpropagation, PyTorch (or TensorFlow/Keras), CNNs, RNN/LSTM, regularisation, optimisers, transfer learning. Checkpoint: image classifier (CNN) and a text classifier.

Tanglish (Tamil + English) Vilakkam

Neural network from scratch, backpropagation, PyTorch/TensorFlow, CNN, RNN/LSTM, optimiser, transfer learning. Checkpoint: image classifier & text classifier.

Learn more
YouTube search: deep learning pytorch full course

6. Phase 5 - Specialisation: NLP / Vision / RL / Time Series (Weeks 27-38) [Advanced]

Explanation (English)
Pick one or two tracks. NLP: tokenisation, embeddings, transformers. Vision: detection, segmentation, GANs/diffusion. RL: Q-learning, policy gradients. Time series: ARIMA, Prophet, LSTM. Checkpoint: one end-to-end portfolio project per track.

Tanglish (Tamil + English) Vilakkam
Onnu rendu track select pannunga. NLP, Vision, RL, Time Series. Ovvoru track-kum oru end-to-end portfolio project pannunga.

Learn more
YouTube search: nlp computer vision reinforcement learning roadmap

7. Phase 6 - LLMs and Generative AI (Weeks 39-46) [Advanced]

Explanation (English)
Transformer internals, pre-training vs fine-tuning, LoRA/QLoRA, prompt engineering, RAG with vector databases, agents/tool-use, evaluation, safety. Checkpoint: a RAG chatbot over your own PDFs and a fine-tuned small model.

Tanglish (Tamil + English) Vilakkam
Transformer internals, fine-tuning, LoRA, prompt engineering, RAG, vector DB, agents, evaluation, safety. Checkpoint: unga own PDF-a vachu RAG chatbot, oru small model fine-tune.

Learn more
YouTube search: llm generative ai roadmap rag fine tuning

8. Phase 7 - MLOps and Deployment (Weeks 47-52) [Advanced]

Explanation (English)
Git, Docker, FastAPI/Flask, experiment tracking (MLflow/W&B), CI/CD, model serving, monitoring, cloud basics (AWS/GCP/Azure), data versioning (DVC). Checkpoint: deploy a model as an API on cloud with monitoring.

Tanglish (Tamil + English) Vilakkam
Git, Docker, FastAPI, MLflow, CI/CD, monitoring, cloud basics. Checkpoint: oru model-a API-a cloud-la deploy pannunga.

Learn more
YouTube search: mlops full course

9. Phase 8 - Mastery: Research, Kaggle, Open Source, Interviews (ongoing) [Advanced]

Explanation (English)
Read papers (arXiv), reproduce results, contribute to open-source (scikit-learn, Hugging Face), compete on Kaggle, write blogs, prepare for interviews (ML system design, coding, statistics). Build a public portfolio.

Tanglish (Tamil + English) Vilakkam
Paper padinga (arXiv), reproduce pannunga, open-source contribute pannunga, Kaggle compete pannunga, blog ezhudhunga, interview prepare pannunga. Public portfolio build pannunga.

Learn more
YouTube search: how to read research papers machine learning

10. Career paths in AI/ML [Beginner]

Explanation (English)

Data Analyst, Data Scientist, ML Engineer, MLOps Engineer, Deep Learning/Research Engineer, NLP Engineer, Computer Vision Engineer, AI/LLM Engineer, Data Engineer, AI Product Manager, Research Scientist. Each needs a different depth in maths vs engineering.

Tanglish (Tamil + English) Vilakkam

Data Analyst, Data Scientist, ML Engineer, MLOps, Research Scientist, NLP/CV Engineer, LLM Engineer... ovvonnukkum maths vs engineering depth maarum. Unga interest paathu select pannunga.

Learn more

YouTube search: ai ml career paths data scientist vs ml engineer

11. Study habits: 70% practice, 30% theory [Beginner]

Explanation (English)

Code every concept, keep a GitHub repo, teach others (blog/video), work on projects with real, messy data, set weekly goals, use spaced repetition for maths formulas, and do not skip fundamentals.

Tanglish (Tamil + English) Vilakkam

Theory 30%, practice 70%. Ovvoru concept-um code pannunga, GitHub-la vaiyunga, matravangalukku solli kudunga, real messy data-la project pannunga, fundamentals skip pannadheenga.

Learn more

YouTube search: how to study machine learning effectively